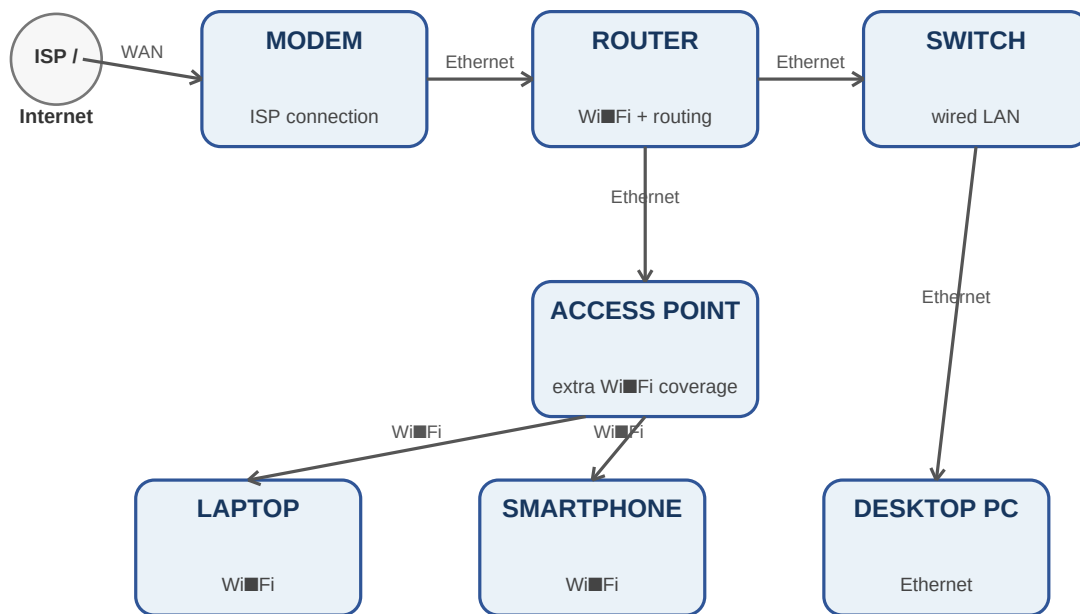


Home Wi-Fi & Networking Hardware

Assignment Answers

1. Labeled diagram of a typical home Wi-Fi setup



Typical home network: Internet → modem → router → switch/access point → devices

How it works: The modem connects the home to the Internet service provider (ISP). The router directs traffic and normally provides Wi-Fi. A switch adds wired Ethernet ports, while an access point provides additional wireless coverage. Laptops and smartphones can connect wirelessly.

2. Main function and real-life example of each device

Device	Main function	Real-life example
Modem	Connects the local network to the ISP service and converts/terminates the cable/DSL/fiber signal to Ethernet	Airmax, the cable/DSL/fiber modem, ONT connects the home network to the Internet
Router	Connects different networks, directs IP packets, assigns local IP addresses (DHCP), and implements NAT with port forwarding	Home routers (e.g., MikroTik, Asus, TP-Link)
Switch	Connects multiple wired devices on the same local network and forwards frames to the appropriate port	A small 8-port switch in a home network, a 24-port switch in a server room
Access Point (AP)	Provides Wi-Fi so wireless devices can join the wired local network	A single AP in a home, multiple APs in a large office provides Wi-Fi coverage
Network cable	Provides a physical path for carrying network signals between two devices	A Cat5e cable can connect a PC to a switch or a router

Note: In many modern homes, one device may combine modem/ONT, router, switch and Wi-Fi access-point functions. The assignment separates them to show the role of each networking component.

3. Three common types of network cable

The illustrations below are simplified diagrams for clarity. The source links point to online reference images on Wikimedia Commons; the cited files are available under the licenses stated on their source pages.

Ethernet (twisted-pair / Cat5e or Cat6)



Twisted-pair Ethernet cable with RJ45/8P8C connector

Commonly used for wired LAN connections such as connecting a PC, printer, switch, router or access point.

Source link: https://commons.wikimedia.org/wiki/File:RJ45_Ethernet_Cable.jpg

Wikimedia Commons — “RJ45 Ethernet Cable.jpg” by Khairil Yusof; CC BY 2.0.

Coaxial cable



Coaxial cable: central conductor, insulation, shield and outer jacket

Commonly used for cable-TV/broadband connections and some RF networking applications.

Source link: https://commons.wikimedia.org/wiki/Category:Coaxial_cables

Wikimedia Commons — Coaxial cables category / reference images.

Fiber-optic cable



Fiber-optic cable: carries data as pulses of light through optical fiber

Commonly used for high-speed Internet backbones, ISP access links, data centers and long-distance communications.

Source link: https://commons.wikimedia.org/wiki/File:Fiber_Optical_Cable.jpg

Wikimedia Commons — “Fiber Optical Cable.jpg” by Bquast; CC0.

4. Router vs. Switch — comparison table

Difference	Router	Switch
Main use	Connects different networks and forwards IP packets	Connects devices within the same LAN and forwards Ethernet frames
Typical location	Usually at the network edge of a home or small office	Usually in the ISP LAN connected to other devices, PCs, printers, servers
Data handled	Primarily makes forwarding decisions using IP addresses	Primarily makes forwarding decisions using MAC addresses and its
Ports / connections	Often has a WAN/Internet port plus several LAN ports	Usually has many Ethernet ports for wired devices; a switch normally
Broadcast domains	Separates networks/broadcast domains and can provide DHCP	Usually keeps hosts on the same LAN/broadcast domain
Wireless capability	Home routers commonly include built-in Wi-Fi.	A standard Ethernet switch is wired; Wi-Fi requires a separate access

Quick summary: A router connects networks and decides where IP traffic should go; a switch connects devices inside a LAN and forwards Ethernet frames to the correct local port.

References / image source links

Ethernet cable image: Wikimedia Commons: RJ45 Ethernet Cable.jpg

https://commons.wikimedia.org/wiki/File:RJ45_Ethernet_Cable.jpg

Coaxial cable images: Wikimedia Commons: Category — Coaxial cables

https://commons.wikimedia.org/wiki/Category:Coaxial_cables

Fiber-optic cable image: Wikimedia Commons: Fiber Optical Cable.jpg

https://commons.wikimedia.org/wiki/File:Fiber_Optical_Cable.jpg

Prepared as an assignment answer sheet. The network diagram and cable drawings in this PDF are original simplified illustrations; the online links are provided as image/source references as requested.